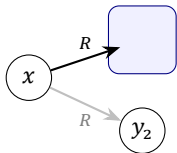


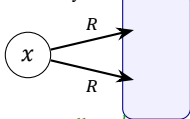
$\exists R.C$   
 “at least one  $R$ -link to  $C$ ”



$x \in (\exists R.C)^I \quad \square$

There exists  $y_1 \in C^I$  with  
 $(x, y_1) \in R^I$

$\forall R.C$   
 “every  $R$ -link to  $C$ ”



no  $R$ -link

$z \in (\forall R.C)^I$  (vacuous)

**Distinction:** Absence in RDF data  $\neq$  absence in an interpretation.  
 Under OWA, a missing  $R$  triple for  $z$  does NOT entail  $z : \forall R.C$ .